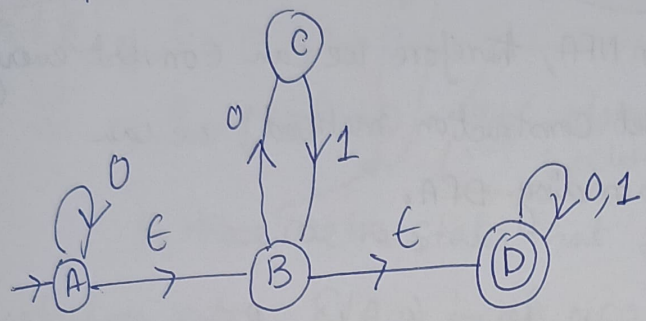
ε-NFA

We will see a special kind of FA or special kind of NFA, called epsilon-NFA (ε-NFA). ε-NFA is different from ordinary NFA, in the sense that, if you look at any transition, that may even contain epsilon (ε). An example of ε-NFA is given below.



An ε-NFA is defined by the five-tuple $(Q, \Sigma, \delta, q_0, F)$ where δ is only different than DFA and NFA.

Here,

$$\delta : Q \times \Sigma \cup \{\epsilon\} \rightarrow 2^Q$$

~~where~~ which means, on a particular state $q_i \in Q$, on seeing some symbol $x \in \Sigma$, we will go to some state(s) or on a particular state $q_i \in Q$, without seeing anything a transition is possible.

for ex: $\delta(A, \epsilon) = B$.

ε-closure:

ε-closure of state A: ε-closure of any state is nothing but what are all the states which you can reach only on seeing ε from that state.

$\therefore$ ε-closure(A) = {A, B, D}

$A \xrightarrow{\epsilon} A$ (always 9 can reach A on seeing ε)

$A \xrightarrow{\epsilon} B$ (A can reach B on seeing ε)

$B \xrightarrow{\epsilon} D$ (B can reach D on seeing ε)

Q How can I convert ϵ-NFA to NFA?

Q We have seen DFA, NFA, and ϵ-NFA. Which one is more powerful?

Ans We have already seen DFA and NFA are equivalent in power.

$$DFA \cong NFA$$

□ Every DFA is nothing but an NFA, therefore we can convert every DFA to NFA, and using "subset construction method", we can convert an NFA to its corresponding DFA.

□ Now, if we remove ϵ from transition function of ϵ-NFA, it converts into an NFA.

$$\delta : Q \times \Sigma \cup \cancel{\{\epsilon\}} \rightarrow 2^Q$$

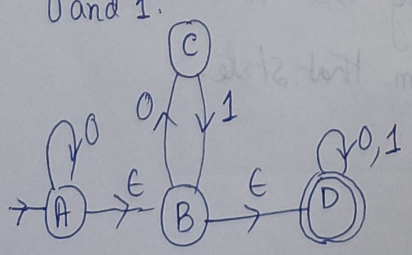
So, conversion from ϵ-NFA to NFA is trivial,

$$DFA \cong NFA \cong \epsilon\text{-NFA}$$

So, all of them are equivalent in power.

□ Conversion of ϵ-NFA to NFA:

Step 1: We will convert the state transition diagram of ϵ-NFA into state transition table. In table, will not have ϵ moves, so we keep only 0 and 1.



	0	1
A	{A,B,C,D}	{D}
B	{C, D}	{D}
C	Φ	{B,D}
D	{D}	{D}

Step 2: To find, what happens when state "A" sees a "0" directly, we first find ϵ-closure(A) ~~A~~ ϵ-closure of A, then apply 0, then again apply ϵ-closure on that result. i.e.;

$$\epsilon\text{-closure}(\delta(\epsilon\text{-closure}(A), 0))$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 0 & & \epsilon^* & \\
A \arrow[r, "\epsilon"] \arrow[dr, "\epsilon"] \arrow[ddr, "\epsilon"] & A \arrow[r, "0"] & A \arrow[r, "\epsilon"] & A & & & \\
 & B \arrow[r, "0"] & C \arrow[r, "\epsilon"] & C & & & \\
 & D \arrow[r, "0"] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

Note: ϵ-closure of C = C only
ϵ-closure of D = D only

So, these are the states that should be present in this particular entry $\delta(A,0)$ for this NFA.

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 1 & & \epsilon^* & \\
A \arrow[r, "\epsilon"] \arrow[dr, "\epsilon"] \arrow[ddr, "\epsilon"] & A \arrow[r] & \Phi \arrow[r, "\epsilon"] & \Phi & & & \\
 & B \arrow[r] & \Phi \arrow[r, "\epsilon"] & \Phi & & & \\
 & D \arrow[r] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 0 & & \epsilon^* & \\
B \arrow[r, "\epsilon"] \arrow[dr, "\epsilon"] & B \arrow[r, "0"] & C \arrow[r, "\epsilon"] & C & & & \\
 & D \arrow[r, "0"] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 1 & & \epsilon^* & \\
B \arrow[r, "\epsilon"] \arrow[dr, "\epsilon"] & B \arrow[r, "1"] & \Phi \arrow[r, "\epsilon"] & \Phi & & & \\
 & D \arrow[r, "1"] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 0 & & \epsilon^* & \\
C \arrow[r, "\epsilon"] & C \arrow[r, "0"] & \Phi \arrow[r, "\epsilon"] & \Phi & & &
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 1 & & \epsilon^* & \\
C \arrow[r, "\epsilon"] & C \arrow[r, "1"] & B \arrow[r, "\epsilon"] \arrow[dr, "\epsilon"] & B & & & \\
 & & & D & & &
\end{tikzcd}$$

$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 0 & & \epsilon^* & \\
D \arrow[r, "\epsilon"] & D \arrow[r, "0"] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

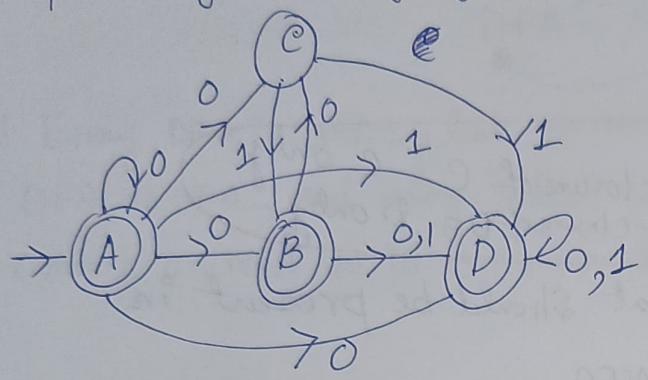
$$\begin{tikzcd}[column sep=small]
 & \epsilon^* & & 1 & & \epsilon^* & \\
D \arrow[r, "\epsilon"] & D \arrow[r, "1"] & D \arrow[r, "\epsilon"] & D & & &
\end{tikzcd}$$

Note: In case of NFA to DFA conversion, no. of states increases
In case of ϵ-NFA to NFA " , no of states remains the same

Step 3:

Initial state is A,

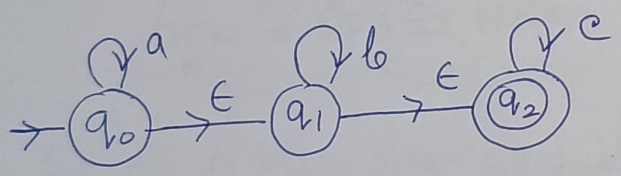
But in the case of final state, we will have, A, B, D as final states, because all the states which can reach final state (D) only upon seeing ϵ are going to be final states.



final NFA from ϵ-NFA.

ϵ-NFA to DFA Conversion

Ex: 1



δ_E	a	b	c	ϵ
→ q_0	q_0	Φ	Φ	q_1
q_1	Φ	q_1	Φ	q_2
* q_2	Φ	Φ	q_2	Φ

Step 1: find out ϵ-closure of every states (here q_0, q_1, q_2)

ϵ-closure(q_0)
$= \{q_0, q_1, q_2\}$

ϵ-closure$(q_1) = \{q_1, q_2\}$

ϵ-closure$(q_2) = \{q_2\}$

δ_D	a	b	c
$*\{q_0, q_1, q_2\}$	$\{q_0, q_1, q_2\}$	$\{q_1, q_2\}$	$\{q_2\}$
$*\{q_1, q_2\}$	Φ	$\{q_1, q_2\}$	$\{q_2\}$
$*\{q_2\}$	Φ	Φ	$\{q_2\}$
Φ	Φ	Φ	Φ

Step2: find out table of DFA from table of ϵ-NFA, where starting state of DFA will be ϵ-closure of starting state of ϵ-NFA i.e, $\epsilon\text{-closure}(q_0) = \{q_0, q_1, q_2\}$

Step3: How to find $\delta_D(\epsilon\text{-closure}(q_0), a)$

Ans Use the following formula

$$\delta_D(\{q_0, q_1, q_2\}, a)$$

$$= \epsilon\text{-closure}(\delta_E(\{q_0, q_1, q_2\}, a))$$

$$\cancel{= \epsilon\text{-closure}(\delta_E(\{q_0\}, a) \cup \delta_E(\{q_1\}, a) \cup (\quad))}$$

$$\cancel{= \epsilon\text{-closure}(q_0 \cup \Phi \cup \Phi)} = \epsilon\text{-closure}(\delta_E(q_0, a) \cup \delta_E(q_1, a) \cup \delta_E(q_2, a))$$

$$\cancel{= \epsilon\text{-closure}(q_0)}$$

$$\cancel{= \{q_0, q_1, q_2\}} = \epsilon\text{-closure}(q_0 \cup \Phi \cup \Phi)$$

$$= \epsilon\text{-closure}(q_0)$$

$$= \{q_0, q_1, q_2\}$$

Step4: $\delta_D(\{q_0, q_1, q_2\}, b) = \epsilon\text{-closure}(\delta_E(\{q_0, q_1, q_2\}, b))$

$$= \epsilon\text{-closure}(\delta_E(q_0, b) \cup \delta_E(q_1, b) \cup \delta_E(q_2, b))$$

$$= \epsilon\text{-closure}(\Phi \cup q_1 \cup \Phi) = \epsilon\text{-closure}(q_1)$$

$$= \{q_1, q_2\}$$

Step5: $\delta_D(\{q_0, q_1, q_2\}, c) = \epsilon\text{-closure}(\delta_E(\{q_0, q_1, q_2\}), c))$

$$= \epsilon\text{-closure}(\delta_E(q_0, c) \cup \delta_E(q_1, c) \cup \delta_E(q_2, c))$$

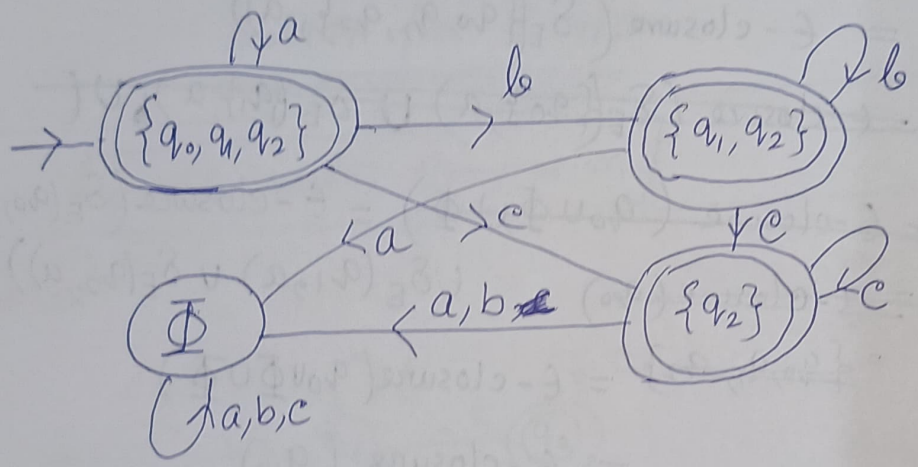
$$= \epsilon\text{-closure}(\Phi \cup \Phi \cup q_2) = \epsilon\text{-closure}(q_2) = \{q_2\}$$

Step 6: follow the same steps for remaining states.

Note: When the first row of state transition table of DFA is generated, we get 2 new states $\{q_1, q_2\}$ and $\{q_2\}$; hence we add these 2 new states at the second and third row and then performing state transition from each of these new states. **Ex:** $\delta_D(\{q_1, q_2\}, a)$, $\delta_D(\{q_1, q_2\}, b)$, $\delta_D(\{q_1, q_2\}, c)$ etc.

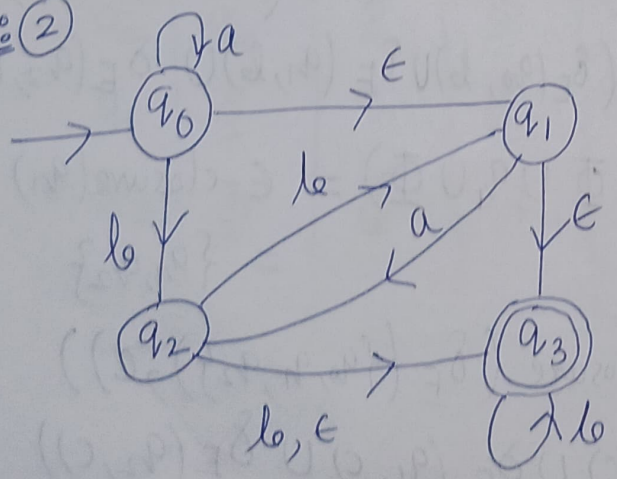
When there is no more new states, we stop the process.

Step 7: Draw the state transition diagram of the DFA, where all the states containing q_2 will become final states.



State transition diagram of DFA

Ex: 2



Step I: ϵ-closure$(q_0) = \{q_0, q_1, q_3\}$

ϵ-closure$(q_1) = \{q_1, q_3\}$

ϵ-closure$(q_2) = \{q_2, q_3\}$

ϵ-closure$(q_3) = \{q_3\}$

Step II: Draw the state-transition table of ϵ-NFA

δ_E	a	b	ϵ
$\rightarrow q_0$	q_0	q_2	q_1
q_1	q_2	Φ	q_3
q_2	Φ	q_1, q_3	q_3
$* q_3$	Φ	q_3	Φ

Step III:

Draw the state-transition table of DFA, where initial state of DFA is ϵ-closure of q_0 i.e, $\{q_0, q_1, q_2\}$

δ_D	a	b
$*\rightarrow\{q_0, q_1, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_2, q_3\}$
$*\{q_0, q_1, q_2, q_3\}$	$\{q_0, q_1, q_2, q_3\}$	$\{q_1, q_2, q_3\}$
$*\{q_2, q_3\}$	Φ	$\{q_1, q_3\}$ ✓
$*\{q_1, q_2, q_3\}$	$\{q_2, q_3\}$	$\{q_1, q_3\}$
$*\{q_1, q_3\}$	$\{q_2, q_3\}$	$\{q_3\}$ ✓
Φ	Φ	Φ
$*\{q_3\}$	Φ	$\{q_3\}$

$\delta_D(\{q_0, q_1, q_3\}, a)$

$= \epsilon\text{-closure}(\delta_E(\{q_0, q_1, q_3\}, a))$

$= \epsilon\text{-closure}(\delta_E(q_0, a) \cup \delta_E(q_1, a) \cup \delta_E(q_3, a))$

$= \epsilon\text{-closure}(q_0 \cup q_2 \cup \Phi)$

$= \epsilon\text{-closure}(\{q_0, q_2\})$

$= \epsilon\text{-closure}(q_0) \cup \epsilon\text{-closure}(q_2)$

$= \{q_0, q_1, q_2\} \cup \{q_2, q_3\}$

$= \{q_0, q_1, q_2, q_3\}$

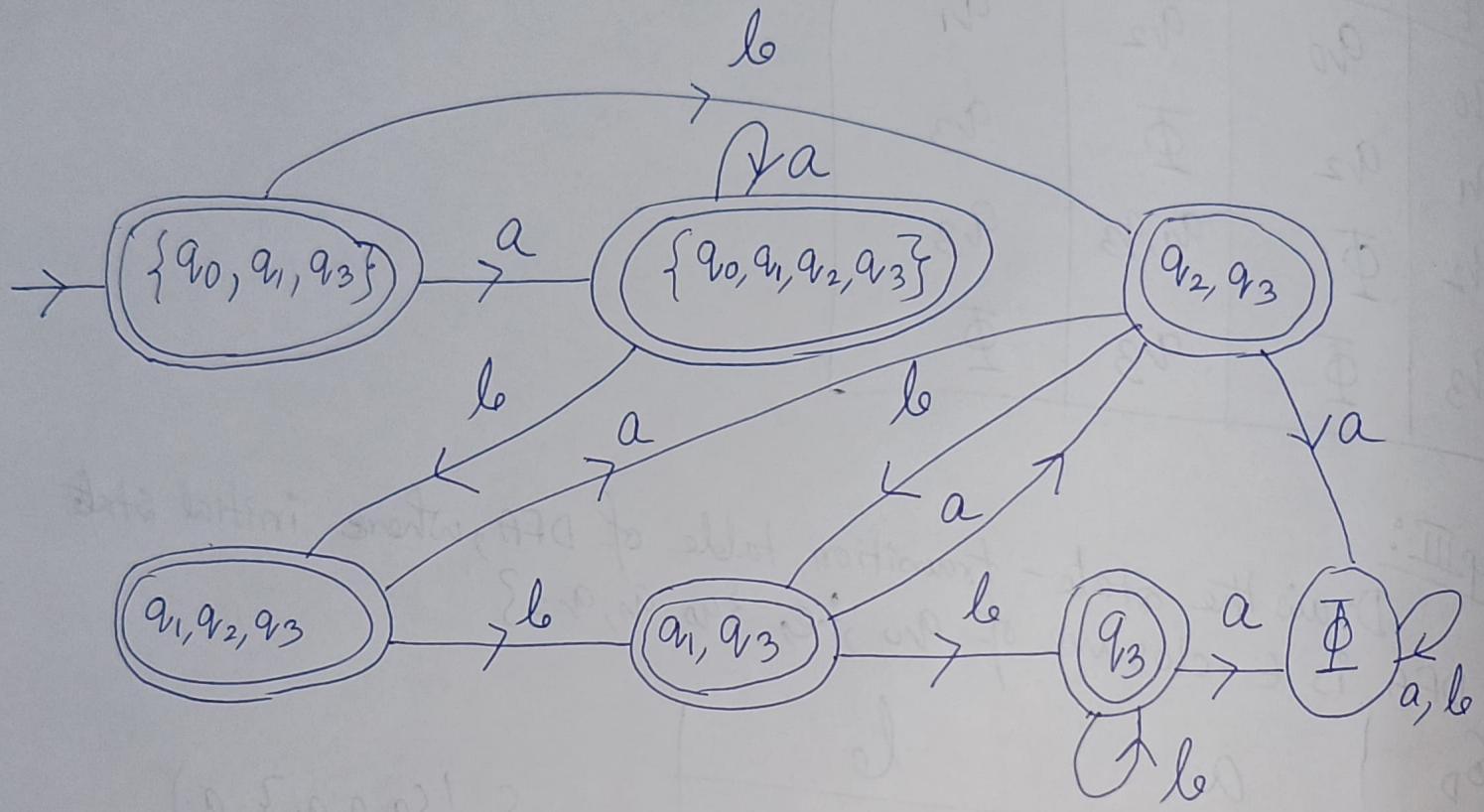
$\delta_D(\{q_0, q_1, q_3\}, b) = \epsilon\text{-closure}(\delta_E(\{q_0, q_1, q_3\}), b)$

$= \epsilon\text{-closure}(\delta_E(q_0, b) \cup \delta_E(q_1, b) \cup \delta_E(q_3, b)) = \epsilon\text{-closure}(q_2 \cup \Phi \cup q_3)$

$= \epsilon\text{-closure}(\{q_2, q_3\}) = \epsilon\text{-closure}(q_2) \cup \epsilon\text{-closure}(q_3)$

$= \{q_2, q_3\} \cup \{q_3\} = \{q_2, q_3\}$

Step IV draw the DFA.



DFA